

CSCE 753 COMPUTER VISION · MODULE 02 · LECTURE NOTES

Cameras and Calibration

What the numbers in a camera matrix mean.

MODULE OUTCOMES

- Write the full projection from world to pixel.
- State what each intrinsic and extrinsic parameter means.
- Explain lens distortion and when it matters.
- Calibrate a camera and assess the result.
- Explain why homogeneous coordinates are used.

1 The projection chain

$$\begin{aligned}
 \mathbf{x}_{\text{cam}} &= \mathbf{R} \mathbf{x}_{\text{world}} + \mathbf{t} && \text{(extrinsics: 6 degrees of freedom)} \\
 (x, y) &= (X/Z, Y/Z) && \text{(perspective division)} \\
 u &= f_x x + c_x, & v &= f_y y + c_y && \text{(intrinsics)} \\
 \mathbf{p} &\approx \mathbf{K} [\mathbf{R} \mid \mathbf{t}] \mathbf{X} && \text{(the whole chain, up to scale)}
 \end{aligned}$$

The perspective division is the only non-linear step and the only lossy one. The rigid transform is invertible, the pixel mapping is invertible, and the division discards Z while keeping X/Z and Y/Z . **Which is why every ambiguity in Module 01 was ultimately a depth ambiguity** — that is literally the quantity the camera throws away, and recovering it is the whole project of Modules 04 through 06.

WHY HOMOGENEOUS COORDINATES

Perspective projection divides by Z, so it is not a linear map and cannot be written as a matrix in ordinary Cartesian coordinates. This is a genuine obstacle rather than a notational inconvenience.

Append a 1 to each point, treat vectors as equivalent up to scale, and the projection becomes linear.

The entire world-to-pixel chain is then a single 3×4 matrix, and composing transformations is matrix multiplication — which is what makes the algebra of the next three modules manageable.

Points at infinity become representable: a point with final coordinate 0 is a direction rather than a location, so **vanishing points become ordinary objects you can compute with** rather than special cases needing separate code. Two parallel lines genuinely do intersect, at an infinite point, and the formula is the same one as for any other pair.

And points and lines become dual. A 2D line is a 3-vector, a point lies on a line exactly when their dot product vanishes, the line through two points is their cross product, and the intersection of two lines is also their cross product. **That duality is what makes Module 04's epipolar algebra tractable** rather than a trigonometric mess.

2 The intrinsics

Parameter	What it is	Typical behaviour
f_x, f_y	Focal length expressed in PIXELS , not millimetres.	Equal unless the pixels are non-square, which is rare on modern sensors. It is (focal length in mm) $\times$ (pixels per mm) , so it scales when you resize the image — halve the resolution and f halves. This is the single most common intrinsics bug.
c_x, c_y	The principal point — where the optical axis meets the sensor.	Near the image centre, displaced by a few pixels to a few tens by sensor mounting tolerance. Fixing it at the exact centre is a reasonable approximation and a measurable error.
s (skew)	Non-perpendicular pixel axes.	Essentially always zero on real sensors. Fix it at zero — estimating it absorbs noise and destabilises the rest.
k_1, k_2, k_3	Radial lens distortion coefficients.	Large on wide-angle lenses, small on narrow ones, and k_1 dominates. Estimating k_3 without corner coverage produces an unstable fit — prefer a two-coefficient model unless the lens is extreme.
p_1, p_2	Tangential distortion, from the lens not being exactly parallel to the sensor.	Small on assembled cameras. Often fixed at zero, and estimating it rarely improves the reprojection error meaningfully.

DISTORTION, AND WHEN TO CARE

Radial distortion bends straight lines — barrel on wide-angle lenses, pincushion on some telephotos — and it is a function of radius from the principal point alone, which is why one or two coefficients suffice.

It is exactly zero at the principal point and largest at the corners. So whether it matters depends on which image regions your features occupy: a method using only central features can reasonably ignore it, and one using the whole frame cannot.

On a wide lens with 10 pixels of corner displacement, ignoring distortion dominates every other error source combined. On a narrow lens it may be well under a pixel and safely neglected. **Measure it rather than assuming either way.**

Undistort once, then work in ideal pinhole coordinates. Every geometric method in Modules 04 through 07 assumes a pinhole camera, and feeding distorted points into the essential matrix estimation produces a plausible-looking, systematically wrong answer — which is far worse than an obvious failure, because nothing signals it.

3 Calibrating a camera

```
# 1. Print a checkerboard. MEASURE the square size with a
# ruler -- "printed at 100%" is frequently false, and a
# 5% scale error becomes a 5% error in every distance
# you later triangulate.
# 2. Capture 15-25 images with the board at VARIED
# orientations, filling different parts of the frame.
# 3. Detect corners to sub-pixel accuracy.
# 4. Solve for K, the distortion coefficients, and each
# board pose by minimising REPROJECTION ERROR -- which
# is Module 05's bundle adjustment in miniature.
# 5. Report the mean reprojection error in pixels.
# Good:          below 0.3 px
# Suspicious:    below 0.05 px (you are overfitting)
# Bad:           above 1 px

# THE THREE FAILURES, in order of frequency:
#
# ALL BOARDS NEAR-FRONTAL
# f and board distance Z are confounded -- a longer
# lens further away fits identically. The residual
# looks excellent and f is wrong. TILT THE BOARD.
#
# AUTOFOCUS LEFT ON
# the intrinsics differ between frames, so no single
# K fits any of them. LOCK FOCUS AND EXPOSURE.
#
# BOARD ONLY NEAR THE IMAGE CENTRE
# distortion is unconstrained where it is largest,
# so k1 is noise. FILL THE CORNERS.
```

A low reprojection error does not mean a correct calibration. It means the model fits the images you supplied, **which is CSCE 633 Module 03's overfitting wearing a geometric costume** — and the frontal-board degeneracy is the clearest case, because it produces an excellent residual and a focal length that may be wrong by tens of percent. **The remedy is the same as it was there: hold data out (§4), and check against something that is not the fit's own residual.**

4 Checking a calibration

- **Undistort an image of a known straight edge** — a door frame, a building, a taped line — and confirm it is straight in the undistorted image. **This catches a wrong k_1 instantly, takes two minutes, and almost nobody does it.**
- **Measure a known distance** by triangulating a pair of points from two calibrated views and comparing against a ruler. **This is the end-to-end check** and it tests the whole chain rather than one parameter.
- **Hold out five calibration images** and report the reprojection error on them without having fitted to them. **CSCE 633's train/test split, applied to geometry** — and a large train/held-out gap is exactly the degeneracy signal of §3.

- **Sanity-check f against the hardware specification:** the lens's focal length in millimetres times the sensor's pixels per millimetre should land within a few percent of your estimate. **A factor-of-two disagreement usually means an image-resize bug (§2).**
- **Recalibrate on a different day with fresh images** and compare the parameters. **Large disagreement indicates that the procedure is unstable rather than that the camera changed**, and the usual cause is insufficient orientation variety.

CALIBRATION DRIFTS

Thermal expansion changes the effective focal length measurably over a camera's warm-up period, which is a matter of minutes — so a calibration captured cold does not describe the same camera an hour later, at precision that matters for metrology.

Mechanical shock moves the sensor relative to the lens, changing the principal point and sometimes the distortion. **A dropped camera is silently decalibrated** and reports nothing about it.

So production systems either recalibrate continuously — self-calibration from scene structure, which Module 05's bundle adjustment supports naturally — **or treat the intrinsics as slowly-varying unknowns** rather than constants.

Which is why Module 13 returns to this. A calibration is a measurement with a validity period, not a constant, and systems built on the assumption that it is constant degrade invisibly — the geometry stays self-consistent and drifts away from the world, which is the hardest class of failure to notice.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Szeliski — Computer Vision, chapter 2 (free PDF)	Image formation, camera models, and distortion — the reference for §1 and §2.
Zhang — A Flexible New Technique for Camera Calibration (free)	The method §3 describes , and what OpenCV implements. Short and worth reading in full.
OpenCV — camera calibration tutorial (free)	The working procedure, with code. Follow it once exactly before deviating.
Hartley & Zisserman — Multiple View Geometry, chapters 2–6	Projective geometry and camera models done properly , including the duality of §1's callout.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Implement the projection chain yourself** and verify it against OpenCV's `projectPoints` on random inputs.
- 2 **Resize an image by half and confirm f halves** by recalibrating on the resized images.
- 3 **Calibrate your own camera** with 20 varied images and report K and the reprojection error.
- 4 **Deliberately calibrate with near-frontal boards only** and compare f against the good calibration. Report the error.
- 5 **Calibrate with autofocus enabled** and report what happens to the residual.
- 6 **Calibrate with the board only in the image centre** and compare k_1 against the good calibration.
- 7 **Undistort a photograph of a straight edge** and measure the residual curvature in pixels.
- 8 **Hold out five images** and report the error on them.
- 9 **Triangulate a known distance** from two views and compare against a ruler measurement.
- 10 **Recalibrate after warming the camera for twenty minutes** and report the change in f .

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Write the full projection chain and say which step loses information.
- 2 Give three reasons homogeneous coordinates are used.
- 3 What are the five intrinsic parameters, and which should be fixed?
- 4 Why does focal length change when you resize an image?
- 5 When does radial distortion matter and when can it be ignored?
- 6 Describe the calibration procedure and name its three common failures.
- 7 Why is a low reprojection error insufficient evidence?
- 8 Give five independent checks on a calibration.